Corporate Finance

Lecture 4: CAPM

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Plan

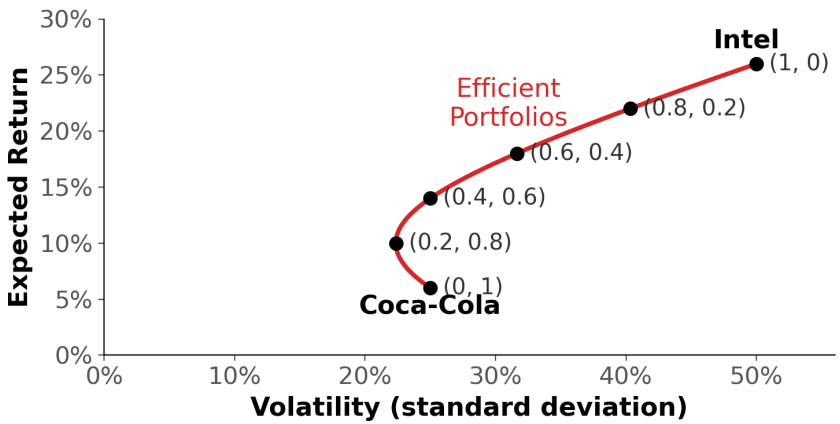
- Recap from lecture 3
- Ask what return a stock must offer to be worth adding to a portfolio: the required return and the beta
- Meet the CAPM: when every investor holds the tangency portfolio, that portfolio is the market itself
- Use the CAPM to determine the risk premium of any stock

Mostly based on the BDM textbook Chap. 11.6-8.

Recap lecture 3

Recap: combining two stocks

- Given two stocks, we choose the **weights** x_1 and x_2, with $x_1 + x_2 = 1$. Each choice is one portfolio, with its own expected return and its own risk.
- Plotted as points, risk across and return up, all the mixes of the two stocks trace a **curve** (Intel and Coca-Cola, zero correlation):



Recap: efficient portfolios and the frontier

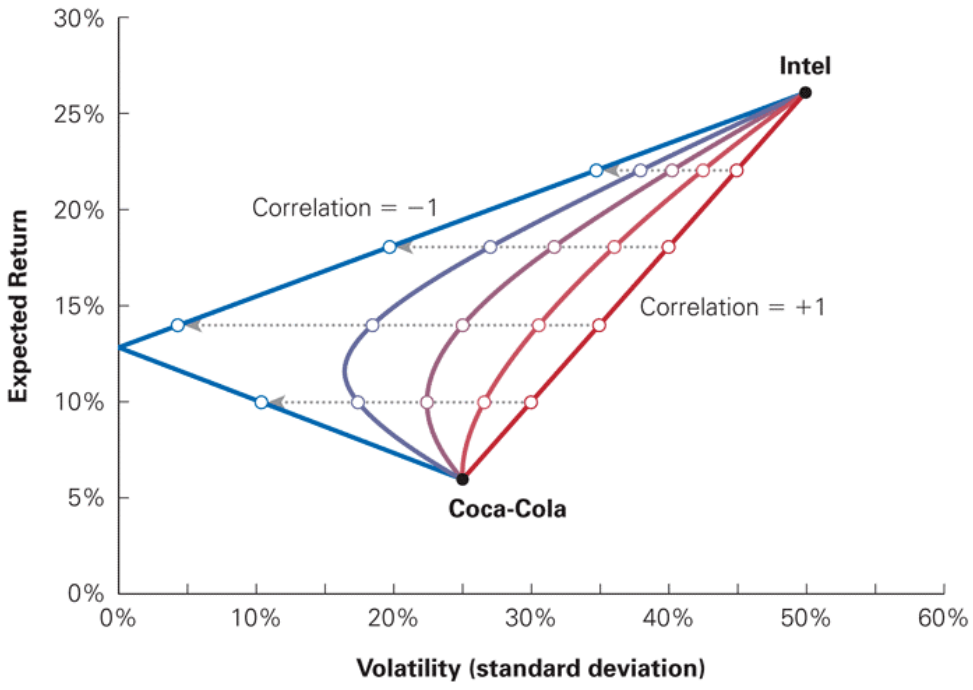
A portfolio is **efficient** when, for its level of risk, no other portfolio gives a higher return; equivalently, for its level of return, no other portfolio has a lower risk.

The **efficient frontier** is the set of all efficient portfolios: the upper part of the curve.

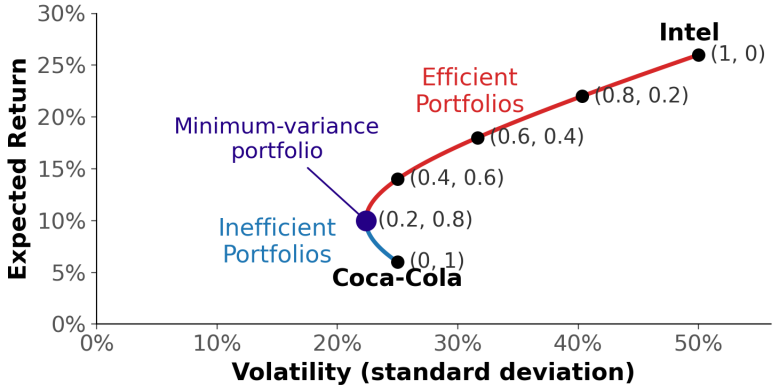
- Example: (0, 1) and (0.4, 0.6) both have 25% volatility, but the second returns 14% against 6%. Holding only Coca-Cola is dominated; the lower part of the curve is inefficient.

Recap: the shape depends on correlation

- **Correlation $+1$**: no diversification. The curve is a straight line between the two stocks.
- **Lower correlation**: the curve bends to the left, the same returns at lower risk.
- **Correlation -1**: the risks can cancel completely; with the right weights the portfolio has zero risk.



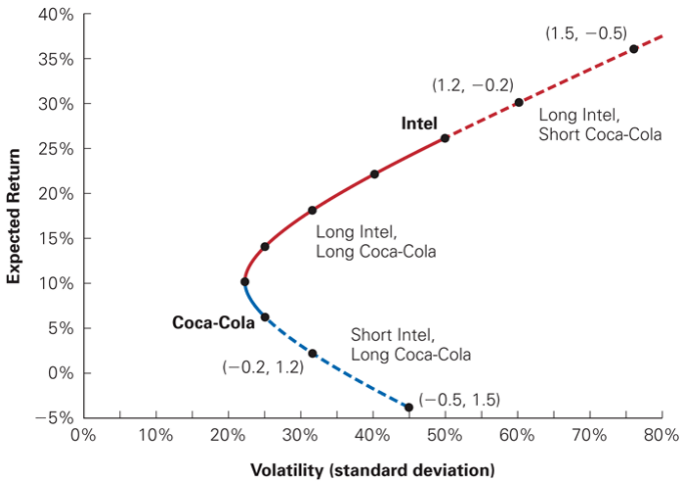
Recap: the minimum-variance portfolio



- The mix with the **lowest risk** of all: the "nose" of the curve, where the efficient set begins. Here (0.2, 0.8), with 22.4% volatility, below either stock alone.
- Its weights depend only on the volatilities and the correlation, **not on the expected returns**.

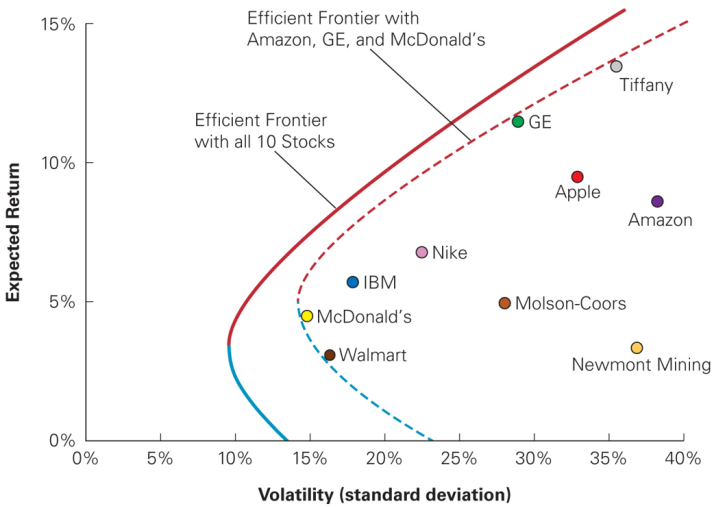
Recap: short sales extend the frontier

- A **short sale** is a negative weight: sell a stock you borrowed, and invest more than your own money in the other one.
- Shorting assets allows to extend the efficient frontier further.



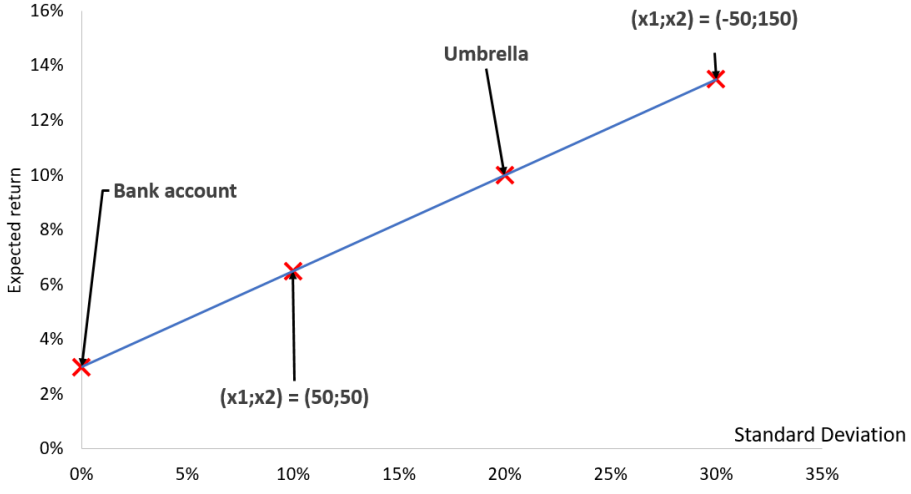
Recap: from two stocks to N

- A third stock turns the curve into a **region** of possible portfolios. The efficient frontier is its **upper-left edge**.
- Every stock we add, even a poor one on its own, moves the frontier further left: more diversification. The best frontier uses all available investments.



Recap: adding the risk-free asset

- The risk-free rate r_f is known in advance: zero volatility, zero correlation with every stock.
- A fraction x in a risky portfolio and $1 - x$ at r_f gives $E[R_p] = r_f + x\,(E[R_U] - r_f)$ and $\sigma_p = x\,\sigma_U$: both grow with x, so every mix is on a **straight line** from the risk-free point. Umbrella (10%, 20%) and a 3% deposit, half and half: 6.5% and 10%.



Recap: borrowing and leverage

- **Leverage** is investing with borrowed money. In the formulas it is a **negative weight on the risk-free asset**, $x_f < 0$, with more than 100% in the risky one.
- Borrow 50 at 3% and put 150 in Umbrella: $x_U = 1.5$, $x_f = -0.5$. Return 13.5% instead of 10%, volatility 30% instead of 20%: leverage scales the risk premium and the risk by the **same factor**.
- It takes you beyond the risky portfolio on the same straight line. Gains and losses are both magnified, so **never judge a return without its risk**.

Recap: the Sharpe ratio

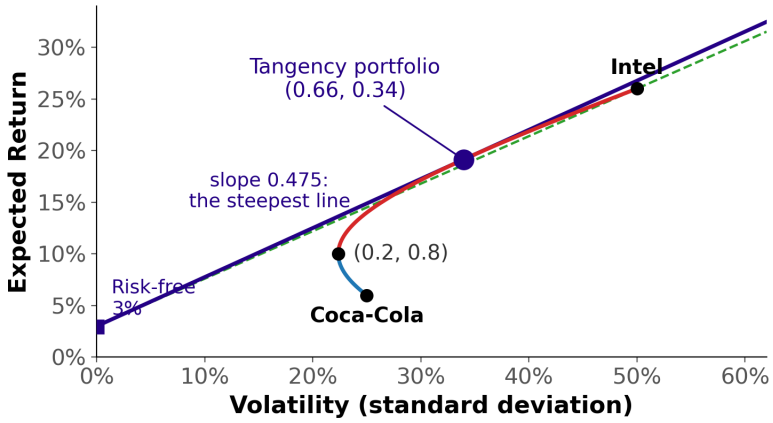
- The slope of the line from the risk-free point is the **Sharpe ratio**: the extra return you get per unit of risk.

$$\text{Sharpe ratio} = \frac{E[R] - r_f}{\sigma}$$

Portfolio	Expected return	Volatility	Sharpe ratio
Coca-Cola (0, 1)	6%	25%	0.12
(0.2, 0.8)	10%	22.4%	0.31
(0.5, 0.5)	16%	28%	0.46
Intel (1, 0)	26%	50%	0.46

- A steeper line pays more at every level of risk, whoever you are. With leverage any return is possible, so **compare Sharpe ratios, not returns**.

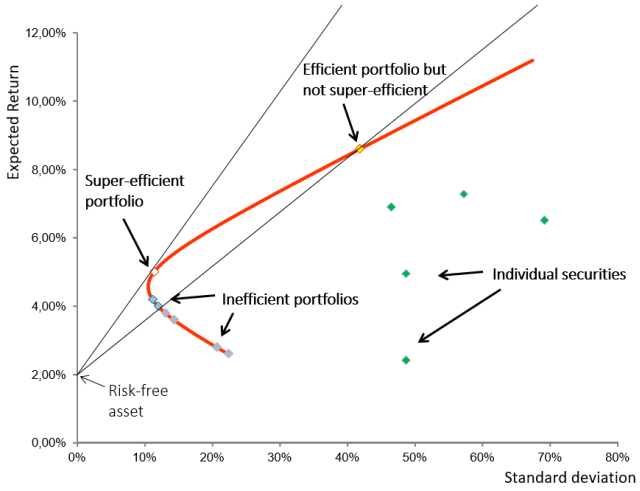
Recap: the Sharpe ratio and the efficient frontier



- The steepest line from the risk-free point is the one **tangent** to the curve. Once a risk-free asset exists, the efficient frontier **is that straight line**:

$$E(R_p) = r_f + \underbrace{\frac{E(R_T) - r_f}{\sigma_T}}_{\text{Sharpe ratio of } T} \times \sigma_p \qquad \text{here } 3\% + 0.475 \times \sigma_p$$

Recap: the super-efficient portfolio

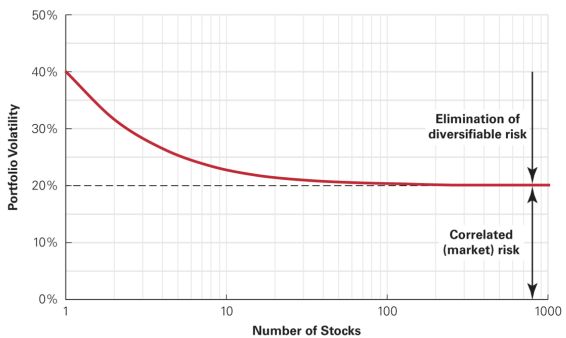


- The tangency point T is the risky portfolio with the **highest Sharpe ratio**: the tangency, or **super-efficient**, portfolio.
- **Everyone holds it.** Risk aversion only sets where you sit on the line: cautious investors lend part of their money at r_f, bold ones borrow and put more than their own money in T (two-fund separation).

Recap: two types of risk

Systematic risk: market-wide (economy, interest rates, inflation). It hits every stock at once, so diversification **cannot** remove it.

Idiosyncratic risk: specific to one firm (its products, its management). It averages out across many stocks, so diversification **does** remove it.



Volatility falls as stocks are added and flattens at the systematic level.

Beginning of lecture 4 content

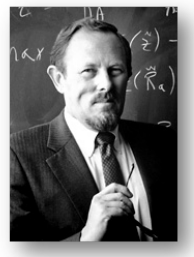
Lecture 3 found the best portfolio of risky assets. Today we ask what that implies for the return each stock has to offer: the CAPM.

Introduction

- Today: how to determine the risk premium a stock must offer, with the **Capital Asset Pricing Model (CAPM)**: Tobin (1958), Treynor (1962), Sharpe (1964), Lintner (1965), Mossin (1966).
- **Why it matters.** The CAPM is the most widely used model linking risk and return. Most firms use it to measure the cost of capital: the discount rate for projects, valuations and regulated prices.



James Tobin



William Sharpe



Harry Markowitz

Outline

A. Required returns: is a stock worth adding to a portfolio? (Chap. 11.6)

B. From the tangency portfolio to the market portfolio (Chap. 11.7)

C. The CAPM, and how to read it (Chap. 11.8)

A. Required returns

- Last class we met the **Sharpe ratio**: the extra return we get for every unit of risk we take. Today we use it to decide whether an asset is **worth adding to a portfolio**.
- You hold a portfolio P. Someone offers you a stock A. **Should you buy some?**

The answer, in one sentence. A stock is worth adding only if its expected return pays for the risk it adds to *your* portfolio. And the risk it adds depends on how it **moves with what you already hold**, not on how risky it is on its own.

- The rest of this section unpacks that sentence in three steps:
  1. measure the risk A adds to P;
  2. measure what that extra risk is worth;
  3. compare with what A actually pays.

Step 1: the risk a stock adds

- Add a small amount δ of A to P, financed by borrowing at r_f. The extra return is easy: $\delta\,(E(R_A) - r_f)$.
- The extra risk is not $\delta\,\sigma_A$. For a small δ the volatility of the portfolio rises by about

$$\Delta\sigma \approx \delta \times \sigma_A \times \rho_{A,P}$$

  δ: the fraction of the portfolio put into A; σ_A: the volatility of A on its own; $\rho_{A,P}$: the correlation between A and the portfolio P.
- The **correlation** is in the formula: a stock that moves with your portfolio adds its full risk, an uncorrelated one almost none, a negatively correlated one **reduces** risk. The same diversification story as Bore Industries.
- Where the formula comes from: the appendix at the end of the deck.

Step 2: what is that extra risk worth?

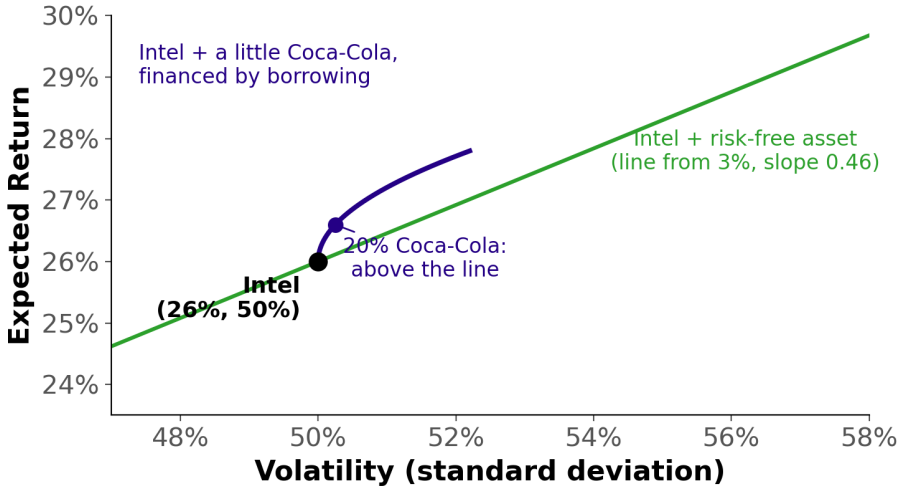
- You could take the same extra risk another way: **leverage**, borrowing to buy more of P.
- Leverage on P pays the Sharpe ratio of P for each unit of risk (Lecture 3). So extra risk $\Delta\sigma$ taken through P would earn

$$\Delta\sigma \times \frac{E(R_P) - r_f}{\sigma_P}$$

  $\Delta\sigma$: the extra risk from step 1; $E(R_P) - r_f$: the risk premium of P; σ_P: its volatility. The fraction is P's Sharpe ratio.
- That is the **benchmark**. A is worth adding when its extra return (step 1, left) is at least what the same risk earns through P (right):

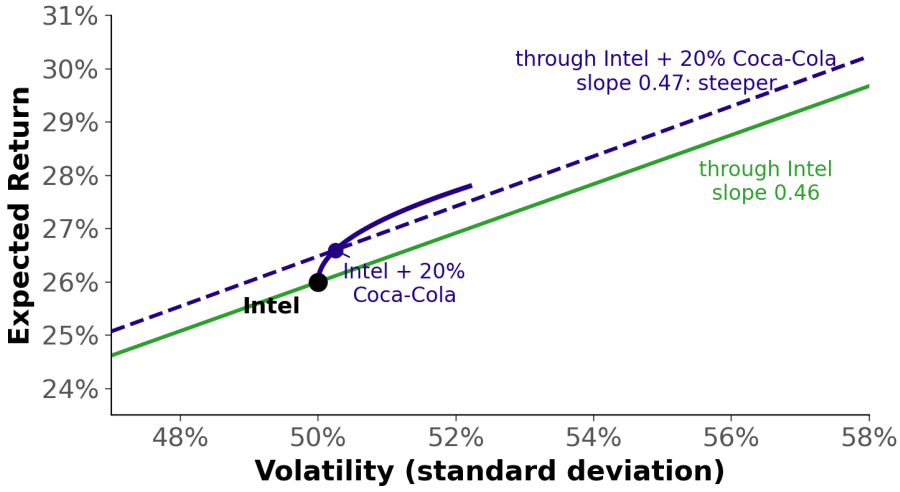
$$\delta\left(E(R_A) - r_f\right) \;\geq\; \Delta\sigma \times \left(E(R_P) - r_f\right)/\sigma_P$$

In a picture: Intel and Coca-Cola



- **Setting.** Your portfolio P is Intel (26%, 50%); the stock A on offer is Coca-Cola (6%, 25%), uncorrelated with Intel; the risk-free rate is 3%. The chart zooms in around Intel.
- **Green line**: Intel mixed with the risk-free asset, the line from Lecture 3 (slope = Intel's Sharpe ratio, 0.46). Beyond Intel it is leverage on Intel.

In a picture: Intel and Coca-Cola (con't)



- **Blue curve**: borrow a little at 3% and put it into Coca-Cola, on top of Intel. The dot is $\delta = 20\%$: return 26.6%, volatility 50.3%.
- **The new line**: draw the line from the risk-free rate through that dot. It is **steeper** than the line through Intel: Sharpe ratio 0.47 against 0.46. Adding a little Coca-Cola **raises the Sharpe ratio**, at every level of risk.

Step 3: from the inequality to the rule

- Start from the inequality of step 2 and put in the extra risk from step 1, $\Delta\sigma = \delta\,\sigma_A\,\rho_{A,P}$:

$$\delta\,(E(R_A) - r_f) \;\geq\; \delta\,\sigma_A\,\rho_{A,P} \times \frac{E(R_P) - r_f}{\sigma_P}$$

- δ appears on both sides, so the size of the purchase does not matter: **cancel it**.

$$E(R_A) - r_f \;\geq\; \sigma_A\,\rho_{A,P} \times \frac{E(R_P) - r_f}{\sigma_P}$$

- Move the volatility of P next to the other risk terms:

$$E(R_A) - r_f \;\geq\; \underbrace{\rho_{A,P}\,\frac{\sigma_A}{\sigma_P}}_{\text{a single number}} \times \big(E(R_P) - r_f\big)$$

Step 3: that single number is the beta

- **Remember the beta** from Lecture 2, the third monster: the covariance of two things divided by the variance of the one that moves. It tells you how much one moves when the other moves by 1%.
- Here the thing that moves is the portfolio P, and we ask how much A moves with it. Using $\text{Cov}(R_A, R_P) = \rho_{A,P}\,\sigma_A\,\sigma_P$:

$$\beta_{A,P} = \frac{\text{Cov}(R_A, R_P)}{\text{Var}(R_P)} = \frac{\rho_{A,P}\,\sigma_A\,\sigma_P}{\sigma_P^2} = \rho_{A,P}\,\frac{\sigma_A}{\sigma_P}$$

  Exactly the single number from the previous slide.
- So the rule reads: buy A if $E(R_A) - r_f \geq \beta_{A,P}\,(E(R_P) - r_f)$. The beta is how we will use comovement from now on: it scales P's risk premium into the premium A must offer.

Step 3: the required return

- Move r_f to the right-hand side. The rule becomes a **required return** r_A:

$$E(R_A) \geq r_A = r_f + \beta_{A,P}\left(E(R_P) - r_f\right)$$

- Read it as: **risk-free rate, plus the portfolio's risk premium scaled by how much A moves with the portfolio**.
  - $\beta_{A,P} = 1$: A must pay the same premium as P.
  - $\beta_{A,P} = 0.5$: half the premium is enough, because A adds only half as much risk.
  - $\beta_{A,P} < 0$: A reduces risk, so it is worth holding even below r_f.
- Buy A if it pays at least r_A; if it pays less, the best move is to short it.

Example: should we add Coca-Cola to Intel?

- You hold Intel: expected return 26%, volatility 50%. Treasury bills: 3%. Coca-Cola: expected return 6%, volatility 25%, correlation with Intel 0.
- Beta of Coca-Cola with respect to Intel: $\beta = 0 \times 25\%/50\% = 0$.
- Required return: $r_{\text{Coca-Cola}} = 3\% + 0 \times (26\% - 3\%) = 3\%$.
- Coca-Cola pays 6%, above 3%: **yes, buy some.** It looks poor next to Intel (a quarter of the return), but being uncorrelated it adds almost no risk, so any return above the risk-free rate is worth having. Lecture 3 agreed: the tangency portfolio holds 34% of Coca-Cola.
- **The twist.** With a correlation of 0.5 instead of 0: $\beta = 0.5 \times 25\%/50\% = 0.25$, required return $3\% + 0.25 \times 23\% = 8.75\%$, above the 6% it pays: **no.** Same stock, same return, same risk; the correlation alone flips the decision.

From the required return to the CAPM

- The rule holds for **any** portfolio P: the required return of a stock is the risk-free rate plus its beta with respect to P times P's risk premium.
- Lecture 3 showed that every investor holds the same risky portfolio, the tangency portfolio. Section B asks what that portfolio must be when everyone holds it, and finds the answer: **the market**.
- Replace P by the market portfolio and the required return becomes the **CAPM**.

B. From the tangency portfolio to the market

- Section A gave a rule that works for **any** portfolio P:

$$r_A = r_f + \beta_{A,P}\left(E(R_P) - r_f\right)$$

- Two questions decide everything that follows:
  1. **Which P does everyone hold?** Lecture 3 said: the tangency portfolio. But which portfolio is that, concretely?
  2. **What does that imply for each stock's return?**
- Where we are going, in four steps: the tangency portfolio turns out to be **the whole market**, and every stock must then pay

$$E(R_i) = r_f + \beta_i\left(E(R_M) - r_f\right) \qquad \text{the CAPM}$$

Step 1: three assumptions

1. Everyone can lend and borrow at the risk-free rate r_f. So the efficient set is the straight line from r_f tangent to the frontier (Lecture 3).

2. Everyone shares the same estimates of expected returns, volatilities and correlations. So everyone computes the **same** frontier and the **same** tangency portfolio.

3. Everyone holds an efficient portfolio. So everyone holds that one tangency portfolio, mixed with the risk-free asset in their own proportion.

Strong assumptions. We look at what they imply first, and at their realism at the end.

Step 2: from tangency portfolio to market portfolio

- Start with Lecture 3. If the world had only Intel and Coca-Cola, everyone would hold the tangency mix: **65.7% Intel, 34.3% Coca-Cola**. The only change between investors would be how they combine it with the risk free asset: the cautious investor holds less of it, the bold one borrows to hold more.
- **Now add up every investor.** Since each one keeps the 65.7 / 34.3 mix, the total money held in Intel is 65.7% of all the money invested in stocks.
- But every share of Intel is held by someone. So the total money held in Intel is also the **market value of Intel**: its price times the number of shares.
- Conclusion: Intel's market value must be 65.7% of the market, and Coca-Cola's 34.3%. **The tangency weights are the market-value weights.**
- The same argument works when adding N stocks: the tangency portfolio holds all stocks, each in proportion to its market value.

Step 2 (con't): what is the market portfolio?

The **market portfolio** M is the portfolio of **all** risky assets, each with a weight equal to its market value divided by the total value of the market.

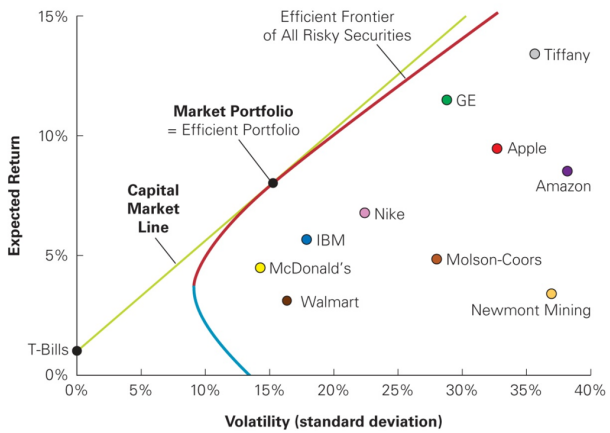
- **Example.** If all stocks together are worth EUR 1 000 billion and a company is worth EUR 60 billion, its weight in the market portfolio is 6%.
- A weight that depends on market value: this is exactly how the value-weighted indices of Lecture 1 are built (CAC 40, S&P 500, MSCI World). **Holding the market** means holding an index fund.

In equilibrium, the tangency (super-efficient) portfolio is the market portfolio.

Step 3: the Capital Market Line

- Everyone mixes the market and the risk-free asset.
- The line that contains all the combinations of the risk free asset and the market portfolio is called the **Capital Market Line** (CML):

$$E(R_p) = r_f + \frac{E(R_M) - r_f}{\sigma_M} \times \sigma_p$$



Step 4: every stock pays its required return

- Take any stock i and ask what happens if it did **not** pay its required return with respect to M.
- **Suppose it paid more.** By section A, adding a little of i to M would raise the Sharpe ratio. Then M would not be the best portfolio, and everyone would be buying i. Its price would rise until the extra return disappears.
- **Suppose it paid less.** Shorting a little of i would raise the Sharpe ratio. Everyone would sell, its price would fall until the gap closes.
- So in equilibrium **no stock can beat or trail its required return**: each pays exactly

$$E(R_i) = r_f + \beta_i \left(E(R_M) - r_f\right)$$

- If every asset must comply this equation, we can use it to evaluate the returns of assets. We say that it is a pricing

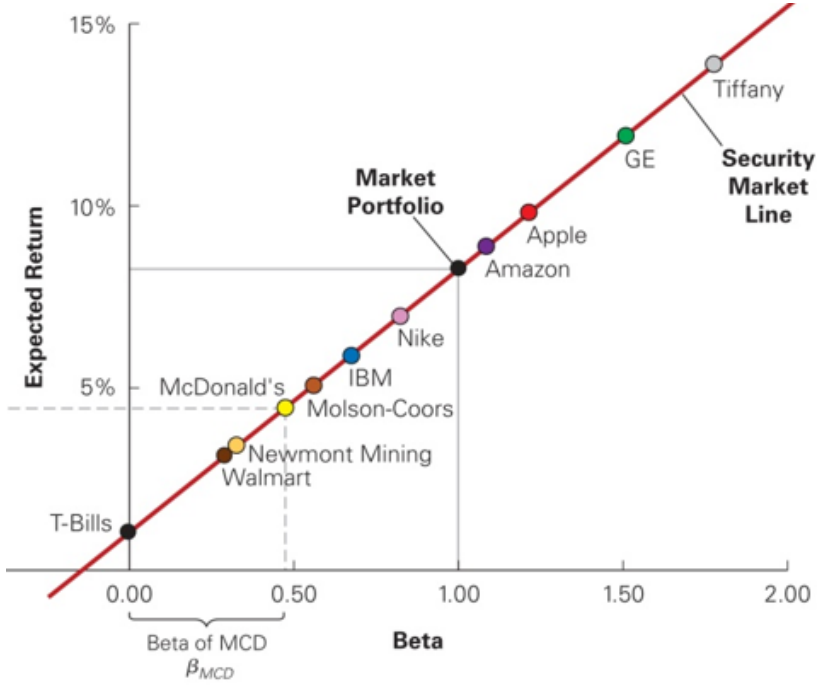
C. The CAPM, and how to read it

$$E(R_i) = r_f + \underbrace{\beta_i \left(E(R_M) - r_f\right)}_{\text{risk premium of stock } i} \qquad \beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)}$$

- β_i is the beta of stock i **with respect to the market**: how much i moves when the market moves by 1%.
- Only the part of i's risk that **moves with the market** is paid for. Its idiosyncratic risk earns nothing: investors can diversify it away for free, so nobody needs to be compensated for it (the two types of risk).
- $\beta_i = 1$: the market's own premium. $\beta_i = 0$: just r_f. $\beta_i < 0$: below r_f, since the stock is insurance against market falls.

The CAPM is a line: the Security Market Line

- Draw the CAPM with **beta** on the horizontal axis and expected return on the vertical one. The formula is a straight line, the **Security Market Line** (SML):
  - at $\beta = 0$ it gives r_f, the risk-free asset;
  - at $\beta = 1$ it gives $E(R_M)$, the market;
  - its **slope** is the market risk premium $E(R_M) - r_f$.
- The CAPM holds for every asset, so **every stock and every portfolio sits on this line**: find its beta, read its expected return.



Example 1: expected return of a stock

- Risk-free rate 4%. Market: expected return 10%, volatility 16%. Stock A: volatility 16%, correlation with the market 0.33.
- **Beta**: $\beta_A = 0.33 \times 16\%/16\% = 0.33$.
- **Expected return under the CAPM**: $E(R_A) = 4\% + 0.33 \times (10\% - 4\%) = 6\%$.
- **The CML portfolio with the same risk.** A 16% volatility on the CML means 100% in the market: expected return 10%.
- Same volatility, 10% against 6%. Stock A carries as much risk as the market, but only a third of it is market risk; the rest is idiosyncratic and earns nothing.
- **On the SML**, Stock A sits at beta 0.33, a third of the way from the risk-free asset to the market.

Example 2: a negative beta

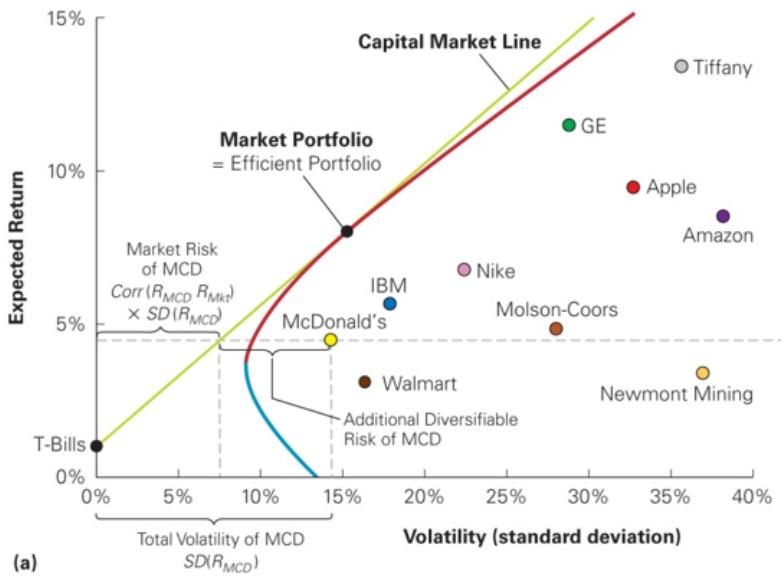
- Stock B has a beta of -0.3. Risk-free rate 4%, expected market return 10%.
- **Expected return under the CAPM**:
  $E(R_B) = 4\% + (-0.3) \times 6\% = 2.2\%$.
- Below the risk-free rate. Why would anyone hold it? Because B **rises when the market falls**: it is insurance for the rest of the portfolio, and investors pay for insurance by accepting a lower return.
- **On the SML**, Stock B sits to the left of the risk-free asset, at beta -0.3, below r_f.

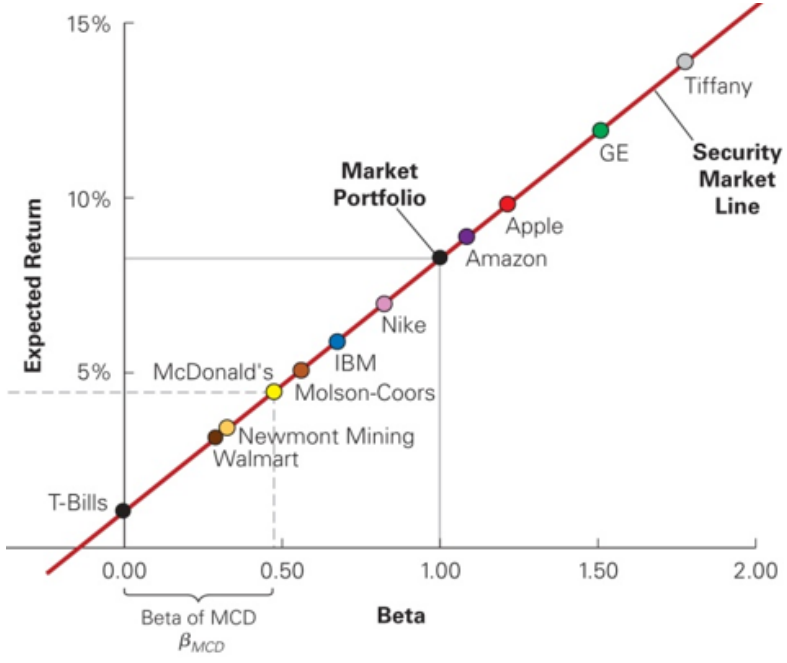
CML versus SML: what is different

	Capital Market Line (CML)	Security Market Line (SML)
Horizontal axis	Volatility (total risk)	Beta (market risk only)
What sits on it	Only mixes of the market and the risk-free asset	Every stock and every portfolio
Slope	Sharpe ratio of the market	Market risk premium $E(R_M) - r_f$
Question it answers	Best return for a given total risk	Return a given asset must pay

- **CML**: where you should invest. It comes from Lecture 3: the efficient frontier once the risk-free asset exists.
- **SML**: what each asset pays. It is the CAPM equation drawn.

CML versus SML: one stock on both





- **Left**: McDonald's lies **below** the CML. Its total volatility is 15%, but only part of it moves with the market; the rest is diversifiable and earns nothing, so the stock pays less than a CML portfolio of the same risk.
- **Right**: the same stock sits **exactly on** the SML at its beta of 0.45. Only market risk is paid, and the SML pays it.

How it all connects

1. **Lecture 3**: combining risky stocks traces the efficient frontier.
2. **Add the risk-free asset**: the best mixes lie on the steepest line from r_f, tangent to the frontier. Everyone holds the tangency portfolio.
3. **Add the three assumptions**: everyone holds the same tangency portfolio, so it must be **the market**.
4. **The CML** is that tangent line with the market as tangency point: where every investor sits, more or less far from r_f depending on their taste for risk.
5. **The CAPM** is what each asset must pay for the market to be the best portfolio: $E(R_i) = r_f + \beta_i \left(E(R_M) - r_f\right)$.
6. **The SML** is the CAPM drawn, with beta on the horizontal axis.

The beta of a portfolio

- A portfolio with weights x_i has return $R_p = x_1 R_1 + x_2 R_2 + ...$ Covariance is linear, so its beta is the **weighted average of the betas**:

$$\beta_p = \frac{\text{Cov}(R_p, R_M)}{\text{Var}(R_M)} = x_1 \beta_1 + x_2 \beta_2 + ...$$

- **Example.** Danone has a beta of 0.5, Ubisoft 1.8. Risk-free rate 4%, expected market return 10%. An equally weighted portfolio:

$$\beta_p = 0.5 \times 0.5 + 0.5 \times 1.8 = 1.15$$

$$E(R_p) = 4\% + 1.15 \times 6\% = 10.9\%$$

Summary: the CAPM formula

$$E(R_i) = r_f + \beta_i \left(E(R_M) - r_f\right)$$

- $E(R_i)$: the **expected return of stock** i, which in equilibrium is also the return investors require to hold it.
- r_f: the **risk-free rate**, what you earn with no risk at all.
- $E(R_M) - r_f$: the **market risk premium**, the extra return the market portfolio pays above the risk-free rate.
- $\beta_i = \text{Cov}(R_i, R_M)/\text{Var}(R_M)$: the **beta** of stock i, how much it moves when the market moves by 1%. It scales the market premium into the premium of stock i.
- In words: **the return of a stock is the risk-free rate plus its share of the market premium, and that share is its beta.**

Summary: what to remember from the CAPM

- **The market portfolio is the only efficient portfolio of risky assets.** The best you can do at any level of risk is to mix it with the risk-free asset: the Capital Market Line.
- **Only market risk is paid.** A stock's premium depends on its beta, its comovement with the market; idiosyncratic risk earns nothing because it can be diversified away. This is the Security Market Line.
- **The assumptions are strong**: everyone shares the same estimates, and everyone can borrow freely at r_f. Neither is true in practice.
- **Yet the CAPM is the standard tool** for the expected return of a stock and the cost of capital of a project, as we saw in the introduction.

Appendix: portfolio volatility approximation

In Section 1 of the slides, I claim that the approximation below holds:

$$\begin{aligned}\Delta\sigma &= \sqrt{\sigma_p^2 + \delta^2\sigma_A^2 + 2\delta\sigma_P\sigma_A\rho_{A,P}} - \sigma_P \\ &\approx \delta\frac{\sigma_P\sigma_A\rho_{A,P}}{\sigma_P}.\end{aligned}$$

To prove this result, define the function f as

$$f(\delta) := \sqrt{\sigma_p^2 + \delta^2\sigma_A^2 + 2\delta\sigma_P\sigma_A\rho_{A,P}}.$$

From Taylor's theorem, it follows that for δ close to zero, we have

$$f(\delta) \approx f(0) + f'(0)\delta.$$

Appendix: portfolio volatility approximation (con't)

Moreover, it is easy to check that

$$f'(\delta) = \frac{\delta\sigma_A^2 + \sigma_P\sigma_A\rho_{A,P}}{\sqrt{\sigma_p^2 + \delta^2\sigma_A^2 + 2\delta\sigma_P\sigma_A\rho_{A,P}}},$$

and, at $\delta = 0$, f and its first derivative simplify to $f(0) = \sigma_P$ and $f'(0) = \frac{\sigma_P\sigma_A\rho_{A,P}}{\sigma_P}$. Thus, Taylor's theorem implies that

$$\sqrt{\sigma_p^2 + \delta^2\sigma_A^2 + 2\delta\sigma_P\sigma_A\rho_{A,P}} \approx \sigma_P + \frac{\sigma_P\sigma_A\rho_{A,P}}{\sigma_P}\delta.$$

The approximation on the previous slide then follows almost immediately.